

Resource Access Protocols

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Abstract—In resource constrained environment of embedded electronic, resource allocation is critical consideration. For tasks to be completed efficiently and within the time frame, protocols are to be followed to allocate resources sufficiently. We know that, in hard real time systems, deadline miss can pose life threatening consequences. The operating system of any real time system follows certain protocols to invoke its kernel to allocate resources on-demand for the executable processes. The seminar paper is a discussion on how the resource protocols are followed inside the OS for successful execution of tasks.

Index Terms—

I. INTRODUCTION TO RESOURCE SHARING IN REAL-TIME SYSTEMS

A new process in real-time system starts share different resources of the system for execution. These resources may vary from data structures, variables, memory (both volatile and non-volatile), files, registers, I/O to processors and others—depending on the task. As stated by [1], “Starting a new process is a heavy job for the OS, which includes allocating memory, creating data structures, and coping code.”. In real-time systems, the process of resource sharing is crucial for ensuring that tasks can execute without unnecessary delays or conflicts and thus meeting their deadlines. Many of the embedded systems with 8-bit processors are widely used in products like dishwashers, microwaves, coffee makers and digital watches; it is well established fact that- “Most embedded systems are constrained in terms of processor speed, memory capacity, and user interface.” [1]. As stated by [1] also, “Real-time embedded systems are often run in highly resource-constrained environments, which make the system design and performance optimization quite challenging.”- thus implying the importance of sharing resources. For allocating resources, required by any task for successful execution, RTOS follows certain protocols.

The protocols can be defined as “a set of rules that govern when and under what conditions each request for resource is granted and how tasks requiring resources are scheduled.” [1]. Whereas, it is also mentioned by [2] that, “Any concurrent operating system(OS) must offer protocols to control the access to resources shared (i.e., a piece of code, such as data structures, I/O devices, buffers, and so on) by competing tasks, thus ensuring mutual exclusion in their respective critical sections (a code segment where shared variables can be accessed)”. To further emphasize the importance, a well-designed access control protocol can be described as

critical rules for preventing deadlocks from occurring in the system [1].

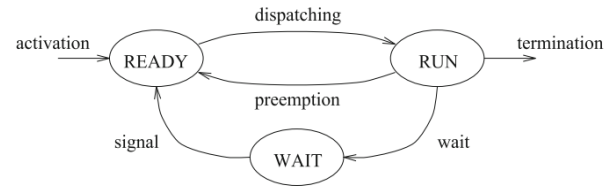


Fig. 1. Waiting state caused by resource constraints [3]

For it's resource constraints and hard deadline requirements, embedded systems often require more stringent resource management compared to general-purpose systems. Hence, “embedded systems must be optimized in terms of size, weight, reliability, performance, cost, and power consumption to fit into the computing environment and perform their tasks” [1]. In general, the RTOS will allocate memory to the process and then loads the module from disk into the memory allocated to it [1]. The process' memory will hold the executable code and initialized data in addition to the uninitialized data and runtime stack into it from the last loaded module with the loader having a default initial size for the stack [1]. Runtime stack can be allocated extra space, not exceeding it's predefined maximum size. [1].

II. COMPARISON OF RESOURCE SHARING PROTOCOLS

Competition for resources in real-time systems can lead to deadline anomalies. RTOS offers shared memories to different processes, tasks or threads to share computational data among themselves. However, these memories must be accessed exclusively and this is the reason, as it was mentioned before, that the memory areas are guarded by mutex or semaphores [1]. The code segment of the critical section of a task allows it to access the shared data [1].

A. Priority Inversion

Task scheduling is greatly affected by resource sharing and subsequently, when a higher priority task is blocked by a lower priority task- it is known as priority inversion [4].

Priority inversion and deadlocks are major common issues contributing to system failure. Although, there is no protocol that can eliminate priority inversion, the goal of access control

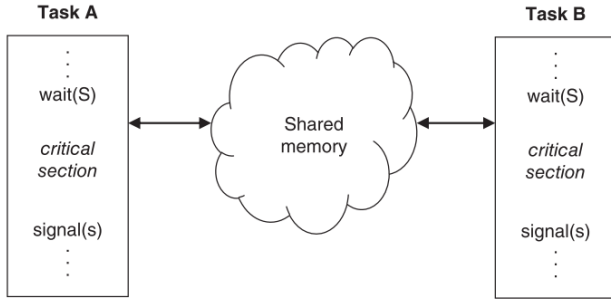


Fig. 2. Shared memory and critical section [1]

is to keep the blocking time of the higher priority tasks under control [1].

For better understanding of how these protocols manage resources allocation among different tasks, a situation can be considered where 3 tasks (J_1 , J_2 and J_3) are running in a real-time system. There, are m type of resources in a system, namely $R_1, R_2, R_3, \dots, R_m$ where each resource comprises η_i number of indistinguishable units of R_i , $i=1, 2, \dots, m$. When a task requests ρ_i units of a resource R_i , request is sent to kernel to lock them and can be denoted as $L(R_i, \rho_i)$. When the resource is longer needed by the task, it is unlocked and can be denoted by $U(R_i, \rho_i)$. As plentiful resources have no effect on scheduling plentiful resources have no effect on scheduling [5], we can avoid considering them here. Also to be considered here- only the serially reusable resources, which are "one that can be used safely by one task at a time and is not depleted by that use" [1]. The examples of these resources are devices, files, databases, and semaphores [1]. The access to resources by every task will be controlled by locking mechanisms.

Binary semaphores- which are also resources, such as mutexes and suspension-based locks, guarantee mutual exclusion [2] by guarding the data shared between several computational tasks [1]. Semaphore is described by [3] as "Operating systems typically provide a general synchronization tool, called a semaphore, that can be used by tasks to build critical sections. A semaphore is a kernel data structure that, apart from initialization, can be accessed only through two kernel primitives, usually called wait and signal." A task which is about to enter a critical section must wait until another task, which is having the required resource, releases it [2]. As stated by [3] that each critical section, operating on a resource R , must start with a wait(W) primitive and end with a signal(S) primitive. All tasks, blocked on a resource with its semaphore, are put in a queue [3]. The overall mechanisms is described by [3] as- "When a running task executes a wait primitive on a locked semaphore, it enters a waiting state, until another task executes a signal primitive that unlocks the semaphore. When a task leaves the waiting state, it does not go in the running state, but in the ready state, so that the CPU can be assigned to the highest-priority task by the scheduling algorithm." [3]. This is how semaphores protect shared resources in real-time

systems by barring concurrent access by multiple tasks.

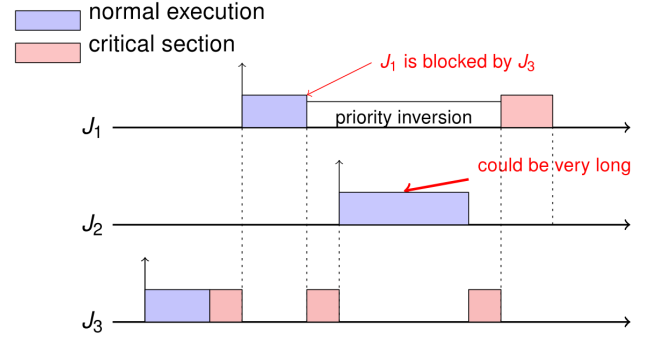


Fig. 3. Priority inversion and its effects [4]

It can be assumed here, for example, that the task J_3 starts at time t_0 . Task J_1 arrives in the system at time t_2 and preempts J_3 while it is executing inside its critical section since J_1 has higher priority. Although, after a while at time t_3 , when J_1 is attempting to start executing at its critical section, it is being blocked on the semaphore and J_3 will continue, again, to execute at its critical section. At time t_4 , yet another task- ' J_2 ' can come at the system, which can preempt J_3 . Subsequently, the blocking time of J_1 is going to prolong one more time. Here, the maximum blocking time of J_1 does depend not only on the length of the critical section executed by J_3 but also on the worst-case execution time of J_2 [3]. Thus, priority inversion is said to occur from the moment of J_1 being blocked by the semaphore until it once again begins executing. This becomes more complex as any task with intermediate priority can preempt J_3 and indirectly blocks J_1 for an indefinite time. This is why, "the blocking time of a task on a busy resource cannot be bounded by the duration of the critical section executed by the lower priority task" [3]. This unbounded priority inversion incident can cause the task, blocked on resource access, miss its deadline [1].

B. Deadlock

Whereas, another phenomenon- deadlock occurs when two or more tasks are waiting for resources to be released by one another, thus creating a cycle of dependencies and situation where none of them can proceed. Deadlock occurs when the following four conditions coincide [1]:

- Mutual exclusion: At least one resource must be held in an exclusive mode.
- Hold and wait: A task will hold a resource while also waiting for another resource.
- No preemption: A resource cannot be forcibly taken from a task holding it.
- Circular wait: There must be a set of tasks where each task waits for a resource held by the next task in the set-creating a circular-chain like dependency issue.

Let it be considered that there are two tasks- T_H and T_L in a system, where T_L starts executing first. After a short

while, it starts using resource A by blocking it. Then, T_H arrives at the system and preempts T_L , as T_H has the higher priority. Later, it continues running by taking the resource B, until attempting to lock the resource A. As A is held by T_L , T_H gets blocked. Similarly, T_L , sometime later, when tries to block resource B, it gets blocked. Now both gets blocked from sharing resources as they are held by each other. No task can make progress in this situation. As it can be seen

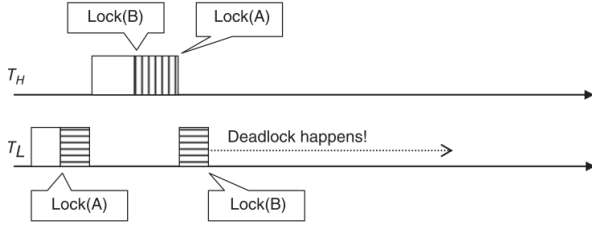


Fig. 4. Deadlock [1]

that resource sharing can cause serious operational problems in real-time systems, there must have to be set of rules to regulate the access to resources by competing tasks. There have been several well-known resource access protocols in place to handle priority inversion and deadlocks efficiently [1]. According to [3], the most common protocols are:

- i. Non-Preemptive Protocol (NPP)
- ii. Highest Locker Priority (HLP), also called Immediate Priority Ceiling (IPC);
- iii. Priority Inheritance Protocol (PIP)
- iv. Priority Ceiling Protocol (PCP)
- v. Stack Resource Policy (SRP)

III. CLASSIC PROTOCOLS TO HANDLE RESOURCE SHARING

It is stated by [3] that the first four protocols, which are mentioned above, have been developed under the fixed priority assignments but some of them have also been extended under the EDF; whereas, the last one is applicable for both fixed and dynamic priority assignments.

A. Non-Preemptive Protocol (NPP)

A simple design to stop unbounded priority inversion, is the implementation of non-preemptive scheduling. Here, the key idea is to make all critical sections nonpreemptable [1]. A task, which locks the resource gets the highest priority among all the other tasks. After executing at its critical section, the task unlocks the resource. As no task is allowed to be preempted while it is holding a resource, this protocol is known as non-preemptive critical section [1]. In non-preemptive scheduling, once a task starts executing, it cannot be interrupted until it completes or voluntarily yields the CPU. This simplifies resource access since tasks complete before others interfere.

- Benefits:
 - No unbounded priority inversion by design.
 - Simple and easy to implement [1].

- No prior knowledge of resource requirements is needed.
- Circular waiting is avoided [1].
- Deadlock is impossible as tasks cannot preempt each other.

• Drawbacks:

- High-priority tasks likely to suffer long blocking times (poor responsiveness)
- Higher priority tasks can be blocked by any lower priority task arbitrarily even when the higher priority task does not need the resource.
- Not suitable for time-critical real-time systems.

This protocol ensures that any higher priority task will be blocked, if needed, at most once [1]. Maximum blocking time is equal to the longest critical section length of all the tasks sharing the resource. This means that the worst case occurs “right after the lower priority task with the longest critical section enters the critical section” [1]. It downgrades the higher priority task to ready and blocked status. According to [1] also, the maximum blocking time of any periodic task is

$$b_i(\text{rc}) = \max\{c_k \mid k = i + 1, i + 2, \dots, n\}$$

where c_k is the longest critical section length of any periodic task T_k . For example, a system with four periodic tasks T_1 , T_2 , T_3 , and T_4 with decreasing order priorities can be considered here. The critical section lengths of the tasks are 5, 4, 7, and 3 respectively.

$$\begin{aligned} \text{Maximum blocking time for } T_1 : b_1(\text{rc}) &= \max\{c_2, c_3, c_4\} \\ &= \max\{4, 7, 3\} = 7 \end{aligned}$$

$$\begin{aligned} \text{Maximum blocking time for } T_2 : b_2(\text{rc}) &= \max\{c_3, c_4\} \\ &= \max\{7, 3\} = 7 \end{aligned}$$

$$\begin{aligned} \text{Maximum blocking time for } T_3 : b_3(\text{rc}) &= \max\{c_4\} \\ &= \max\{3\} = 3 \end{aligned}$$

Task T_4 does not block any other task as it is the lowest priority task. Hence, its maximum blocking time is 0.

B. Priority Inheritance Protocol (PIP)

According to this protocol, a lower priority task will block a higher priority task and attain the higher priority temporarily. This way, the lower priority task will be able to execute its critical section as soon as possible [1]. According to [3], “This prevents medium-priority tasks from preempting τ_i and prolonging the blocking duration experienced by the higher-priority tasks”, where τ_i is a set of periodic tasks. The Priority Inheritance Protocol (PIP) is a classic approach to reduce priority inversion in systems where tasks share mutually exclusive resources (e.g., locks, semaphores). When a low-priority task holds a resource required by a higher-priority task, the low-priority task temporarily inherits the higher priority. This prevents it from being preempted by intermediate-priority

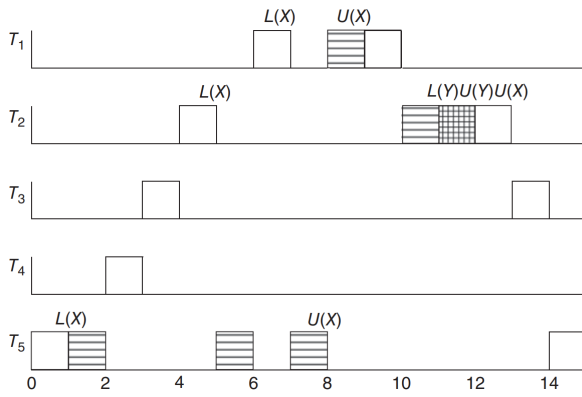


Fig. 5. Schedule of tasks according to PIP [4]

tasks, allowing it to finish quickly and release the resource. In the figure 5, we can see the scheduling of five single-shot task according to PIP protocol.

- T_5 is released at time 0 and start executing.
- At time 1, T_5 locks the resource X that is free. Occupying the resource, T_5 starts executing its critical section.
- At time 2, T_4 comes in the system. Because, it has the higher priority, it will preempt T_5 .
- T_4 completes its execution at time 3 and at the same time T_3 is released and it preempts again T_5 .
- At time 4, T_2 is released and it preempts T_3 . Both T_5 and T_3 are waiting now.
- Now, T_2 attempts to lock X at time 5 which is held by T_5 . So, T_2 gets blocked by T_5 and it inherits T_2 's priority. T_5 starts running causing T_2 and T_3 to wait.
- T_1 is released at time 6 and starts executing by preempting T_5 because of its higher priority. Now, T_2 , T_3 and T_5 are waiting.
- T_1 attempts to lock X at time 7 but the lock is denied because T_5 is holding that. T_5 will inherit T_1 's priority by blocking it and starts executing.
- T_5 completes its critical section execution at time 8 and unlocks X. T_1 now occupies X. However, T_5 still have one unit left to execute.
- T_1 complete its critical section at time 9 and unlocks X. It will continue executing as it has the highest priority in the system.
- At time 10, T_1 complete the execution. T_2 will be granted to lock X.
- At time 11, T_2 locks Y and executes with both X and Y while T_3 and T_5 are waiting.
- At time 12, T_2 unlocks X and Y and continues to execute.
- At time 13, T_3 will start executing after T_2 finishes execution.
- T_3 completes execution and T_5 starts.
- T_5 completes execution at time 15 and thus all tasks are executed.
- Benefits:
 - Reduces unbounded priority inversion [1].

- A task can at most be blocked once by any lower priority task [1].
- It is easy to implement as PIP is supported by most of the commercial RTOSs [1].

According to [1], it was used to solve the famous Mars Pathfinder priority inversion issue in 1997.

• Drawbacks:

- It does not prevent deadlock from happening [1].
- A task might fail to meet its deadline as according to [1], it can be blocked by many lower priority tasks—causing the blocking time to be too long.

C. Priority Ceiling Protocol (PCP)

The Priority Ceiling Protocol (PCP) is an improvement of PIP which assigns a ceiling priority to each shared resource—equal to the highest priority of any task that might access it. A task can only lock a resource if its priority is higher than the ceiling value of any currently locked resource by another task.

• Benefits:

- Prevents formation of deadlocks by denying those resource requests that, if granted, may cause deadlock later [1].
- Eliminates unbounded priority inversion, and also avoids chained blocking [1].

• Drawbacks:

- It entails overhead of the calculation of resource priority ceiling and system priority ceiling, and the overhead of extra context switches due to priority ceiling blockings [1].
- Eventually, a “task that does not need to access resources may suffer at most two context switches due to preemption. A task that accesses resource may suffer additional two context switches incurred by blockings” [1].

D. Stack Resource Policy (SRP)

The Stack Resource Policy (SRP) is a policy which provides stack sharing capacity and simplifies the priority ceiling protocol [1]. The worst-case blocking time of a task is same as PCP under this protocol but this protocol incurs smaller overhead of context switches compared to the priority ceiling protocol [1].

• Benefits:

- No unbounded priority inversion [1].
- No deadlock and no chained blocking [1].

IV. CONCLUSION

The seminar paper is a discussion of the challenges of resource sharing in real-time systems, particularly focusing on priority inversion and deadlocks. It has been discussed that how these issues can lead to significant operational problems, including missed deadlines and system failures. The chapters also provided idea of classic protocols designed to manage resource sharing effectively, such as Non-Preemptive

Protocol (NPP), Priority Inheritance Protocol (PIP), Priority Ceiling Protocol (PCP), and Stack Resource Policy (SRP). Each protocol has its own benefits and drawbacks, making them suitable for different scenarios in real-time systems based upon operations. The discussions highlighted the importance of selecting the appropriate protocol based on the specific requirements and constraints of the given system. By understanding these protocols, developers can better design real-time systems that are robust by ensuring timely and reliable operations.

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